

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}
 D = 1 : & \quad \chi_y(E) = 0, \\
 \text{beginalign*2mm}] D = 2 : & \quad \chi_y(E) = 2(1 + 10y + y^2)(1 - y)^{r-2}, \\
 \text{beginalign*2mm}] D = 3 : & \quad \chi_y(E) = -\chi(E)y(1 + y)(1 - y)^{r-3}, \\
 \text{beginalign*2mm}] D = 4 : & \quad \chi_y(E) = (2 + (2r - 8 - \chi(E))y \\
 & \quad \quad \quad + (8r + 12 - 4\chi(E))y^2 + (2r - 8 - \chi(E))y^3 + 2y^4)(1 - y)^{r-4}, \\
 \text{beginalign*2mm}] D = 5 : & \quad \chi_y(E) = -\chi(E)y(1 + y)(1 + 10y + y^2)(1 - y)^{r-5},
 \end{aligned}$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}
 \mathcal{I}_D(\eta_0, \delta) - \mathcal{I}_D(\eta, 0) &= \mathcal{I}_D(\eta_0, \delta) - \mathcal{I}_D(\eta, \delta) + \mathcal{I}_D(\eta, \delta) - \mathcal{I}_D(\eta, 0) \\
 &\leq \Psi(\eta_0 - \eta, 0) + \Psi(0, \delta) = \Psi(|\eta_0 - \eta|, 0) + \Psi(0, \delta).
 \end{aligned}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}
 \phi_\lambda(v, s') &= f(s') - \lambda_1 \mathbf{d}_{\mathcal{S}_1}(s_1, s'_1)^{p_1} - \lambda_2 \mathbf{d}_{\mathcal{S}_2}(s_2, s'_2)^{p_2} \\
 &\geq B + (B - \lambda_1) \mathbf{d}_{\mathcal{S}_1}(s_1, s'_1)^{p_1} + (B - \lambda_2) \mathbf{d}_{\mathcal{S}_2}(s_2, s'_2)^{p_2} \geq B.
 \end{aligned}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}
 S_{\text{IIB}} &= -T_{M_2} l \int d^2 \xi \sqrt{|\tilde{g}_{ij} - \frac{1}{k^2}(\partial_i \varrho + \tilde{B}_i^{(1)})(\partial_j \varrho + \tilde{B}_j^{(1)})|} \\
 &\quad - \frac{T_{M_2} l}{2} \int d^2 \xi \epsilon^{ij} \left[\tilde{B}^{(2)}_{ij} + 2\partial_i \varrho \left(\tilde{A}^{(1)}_i + m\pi\alpha' b_j \right) \right].
 \end{aligned}$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\langle q_{i0} - Q_{i0} \rangle = 0,$$

$$\langle \dot{q}_{i0} \rangle = 0,$$

$$m\omega_i\omega_j\langle (q_{i0} - Q_{i0})(q_{j0} - Q_{j0}) \rangle = k_B T \delta_{ij},$$

$$m\langle \dot{q}_{i0}\dot{q}_{j0} \rangle = k_B T \delta_{ij},$$

$$\langle \dot{q}_{i0}(q_{j0} - Q_{j0}) \rangle = 0.$$